YIFEI SUN

Principal Software Engineer · GenAI Automation & Backend Platform Architecture · 13 Years

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SUMMARY

Principal-level backend & platform engineer with 13 years of experience building scalable distributed systems and AI-driven automation platforms. Recent work centers on agentic AI infrastructure — designing and shipping an MCP server with tool calling for conversational AI integration, building a GenAI automation platform with RAG-augmented inference, prompt optimization, and multi-provider LLM evaluation, and architecting foundational backend services consumed across the organization. Track record of leading cross-functional platform initiatives spanning 10+ teams, driving technical direction via RFCs, and delivering complex 0→1 systems from inception through production.

LEADERSHIP HIGHLIGHTS

- Designed and shipped agentic AI components: MCP tool-calling server, RAG-augmented inference, prompt evaluation harness
- Defined technical roadmaps and architectural standards adopted org-wide via RFCs and design docs
- Led multi-team platform migrations spanning 10+ engineering teams across 3 departments
- Mentored senior engineers toward Principal-level scope; drove cross-org technical alignment

Technologies: Java, Spring Boot, Python, Node.js · OpenAI/ChatGPT, MCP Protocol, RAG, Prompt Engineering, LLM Orchestration · AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL · GraphQL (Netflix DGS), Microservices, Event-Driven · Datadog, Grafana

INDEPENDENT PROJECT

GenAI Automation Platform & Evaluation System

- Architected an automation platform with hybrid rules + multi-path LLM inference (fast vs. deep), RAG-augmented retrieval over heterogeneous data sources, and a policy overlay — producing explainable, grounded real-time recommendations.
- Built an evaluation harness ("Prompt Lab") for end-to-end simulation, offline backtesting, and prompt/parameter optimization (grid & Bayesian search) with robust observability, hallucination detection, and multi-provider LLM support.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions — enabling safe, iterative automation improvement.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

ChatGPT Apps SDK MCP Server — Conversational AI Integration (0→1)

- Sole backend engineer designing and implementing an MCP server compatible with the OpenAI Apps SDK, exposing CarGurus capabilities as tools callable by conversational AI agents — defining tool schemas, serving infrastructure, and request/response contracts.
- Delivered a production-grade implementation under ambiguity; iterated rapidly with PMs and directors on tool definitions and agent behavior; designed for safe, incremental rollout alongside existing APIs.
- Served as key engineering representative for AI automation strategy, shaping the technical roadmap for LLM/agent integration across the company.

High-Traffic Platform Modernization — Scalable Backend Infrastructure

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core backend behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions).
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes. Produced initial architecture in ~2 weeks; shipped first milestone in 6 months.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

GraphQL Federation — Unified Serving Layer

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, creating a unified serving layer for frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and integration tests ensuring production-grade reliability.

Digital Commerce Platform & Experimentation

- Led backend design and launch of multiple new products using microservice and event-driven architecture; enabled the company to quickly launch and safely retire product experiments without destabilizing core systems.

Cross-Org Platform Leadership & Technical Direction

- Defined technical roadmap and architectural standards for platform services — API design, caching, distributed orchestration — adopted across multiple teams via RFCs and design docs.
- Mentored multiple senior engineers on system design, performance, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Customer Support & Workflow Automation

- Partnered with product, support, and customer-facing teams to modernize internal support workflows and enterprise integrations with clear observability, rollback paths, and operational metrics.

Authentication & Identity Platform

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Platform Initiative

- Drove design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth; designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future platform growth.
- Built initial AWS deployment pipelines and automated testing framework, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners, ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013
B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010